

Problem 15.5

If $a_{\overline{n}|} = x$ and $a_{\overline{2n}|} = y$, express d as a function of x and y .

Solution

We are given

$$a_{\overline{n}|} = x.$$

Using the annuity-immediate formula,

$$x = \frac{1 - v^n}{i}.$$

Also,

$$a_{\overline{2n}|} = y,$$

so

$$y = \frac{1 - v^{2n}}{i}.$$

Since

$$1 - v^{2n} = (1 - v^n)(1 + v^n),$$

we can write

$$y = \frac{(1 - v^n)(1 + v^n)}{i}.$$

But

$$x = \frac{1 - v^n}{i},$$

so

$$y = x(1 + v^n).$$

Therefore,

$$\frac{y}{x} = 1 + v^n.$$

Thus,

$$v^n = \frac{y}{x} - 1 = \frac{y - x}{x}.$$

Now, from

$$x = \frac{1 - v^n}{i},$$

we solve for i :

$$i = \frac{1 - v^n}{x}.$$

Substituting $v^n = (y - x)/x$ gives

$$i = \frac{1 - \frac{y - x}{x}}{x}.$$

Simplifying,

$$i = \frac{2x - y}{x^2}.$$

Finally, the relationship between the effective discount rate d and the effective interest rate i is

$$d = \frac{i}{1 + i}.$$

Substituting $i = (2x - y)/x^2$,

$$d = \frac{\frac{2x - y}{x^2}}{1 + \frac{2x - y}{x^2}}.$$

Therefore,

$$d = \frac{2x - y}{x^2 + 2x - y}.$$

Hence,

$$\boxed{d = \frac{2x - y}{x^2 + 2x - y}}.$$